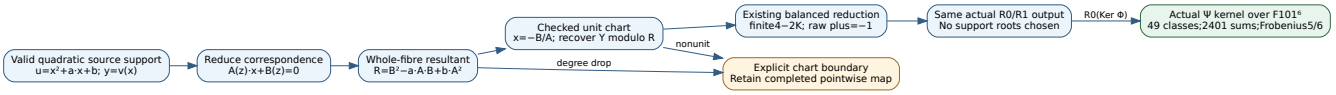
Closed Richelot fibres and the actual Ψ kernel

2026-11-11 · CT23 · unsigned research checkpoint

The new evaluator recovers actual Richelot images without splitting quadratic source support. Its targeted comparison passes192 inputs in each R0/R1 direction. The separate kernel test computes49 actual T0 classes R0(Ker Φ), checks every2401 addition,7-annihilation, geometric dualR0 norm2 and Frobenius eigenvalues5/6 over the existing F101⁶ field. PCC60944e: all four complete12317-class map domains also agree with the old evaluator, including group controls and explicit geometric boundaries. [pcc:unverified]



Exact finite-algebra elimination

For valid u=x²+a·x+b, reduce the existing correspondence g1(x)h1(z)+g2(x)h2(z) to A(z)·x+B(z). Put R=B²−a·A·B+b·A². The identity

(A·x+B)(B−a·A−A·x)+A²u(x)=R

shows R vanishes in the fibre algebra. If R has degree4 and A is a unit moduloR, x=−B/A gives mutually inverse algebra maps between K[z,x]/(u,A·x+B) and K[z]/R. This preserves nilpotents and multiplicities. For u=(x−t)² the resultant is (A·t+B)². No finite-tests-to-global inference is needed for this identity. [pcc:unverified]

If source_scale·target_scale·v(−B/A) is also a unit, recover the target ordinate by the original Richelot equation. Runtime checks source u, the reduced correspondence and the target curve moduloR. The resulting four-point divisor minus2K has raw balanced plus=−1 and uses the unchanged reduction previously used by Cantor addition. Lower-degree supports and infinities reuse the existing completed fibres. [pcc:unverified]

Degree drops and nonunits reject the chart explicitly. Prime inputs retain the old completed map as fallback; extension inputs report a boundary. This does not assert that an unsupported point or map does not exist. [pcc:unverified]

Kernel, proof scope and continuing checks

Actual49-point geometric kernel

The retained global square is Ψ·R0=R1·Φ, with degΨ=49. R0 has2-primary kernel, hence is injective on the49 geometric7-torsion points KerΦ. Their images are49 distinct points killed byΨ and therefore its full geometric kernel. The new chart computes the actual balanced T0 divisor classes in the existing $F101^6$ field. [pcc:unverified]

Native02 checks every image,7-annihilation, all49 geometric dualR0 compositions against[2], all2401 additions and Frobenius5/6. The independent flat-field elliptic route supplies actual7-kernels and checks Φ killing. Full native02/PCC60944e completes the kernel and all-domain comparisons. Earlier partial02/PCCb0738e is preserved unchanged; its scope is not retroactively promoted. [pcc:unverified]

Faithful partial Lean proof

Six ClosedFibres.lean identities pass gate and all local declaration bindings, rootPCCa97f41: quadratic remainder, resultant norm identity, linear/quadratic source recovery under explicit inverse, repeated-support resultant and cleared ordinate recovery. Total source census103, zero trusted native. Full coordinate-ring, scheme/Picard and projective geometry remain the written bridges in CLOSED_FIBRES.md/rootproof0e5bbf. [pcc:unverified]

Primary source: Kunzweiler, DCC92(2024), §3.2 Propositions7–8. Its later degree5/rational-Weierstrass assumptions are not imported as coverage for our split sextics. Existing native/consumer/generic Richelot and balanced engines are reused. KB retrieval is unavailable; no novelty is claimed. [pcc:unverified]

Current scope and next action

The full four-domain12317-class comparison passes exact old/closed equality, both dual2 norms, bijections, paired additions, descent and wrong-model rejection. Each direction has239 lower-degree/12078 quadratic cases and zero fallback. All16 geometric2-classes per direction separately give route counts[1,0,0,2,13]: identity evaluated, two projection and13 ordinate nonunit boundaries. No branch denominator is inverted. PCC60944e. [pcc:unverified]

Integrated complete guard07 passes25native/70reference unit plus three integrations,one old ignored benchmark,all guards/fmt/strictClippy and 109-source/naive/reverse28-inverse74/direct4 controls. PCC1f1423 pins exact source07/receipt. The compact49-class TSV07 is frozen under9efff5. [pcc:unverified]

Independent flatFpR/poly replay02 checks all exported curve-divisor identities,canonical shapes/distinctness and Frobenius5/6,with fixed counts 1/7/1/49 over degrees1/2/3/6. RootPCC4a98a7. It imports no producer tower, closed map or elliptic generator. Group additions and kernel membership retain their separate native/written evidence. [pcc:unverified]

Read CLOSED_FIBRES_STATUS.md, src/closed_fibres.rs and immutable evidence with the closed-fibres prefix. This branch starts readyPR1490/994869034. Parent CI and both frozen-ledger forks remain separate obligations. Next CT24 retains the further nondual7-hop. S3 admission/livebudget block benchmarks and paid work. The current result concerns explicit evaluation and geometric kernel data; development elapsed times do not measure cost. [pcc:unverified]